

Digital On-Demand Tailoring & Garment Delivery Platform for TAILOR24

Build the digital workflow that connects customers, hubs, and independent tailors under a 24-hour delivery promise.

1. Problem Background

TAILOR24 is an on-demand tailoring platform serving customers in Tier-2 / Tier-3 Indian cities. Customers book garment-stitching orders (shirts, trousers, kurtas, sarees, suits) by phone, web, or PWA. Each order is fulfilled through a local **hub** — a physical workshop that runs cutting, stitching, QC, ironing, packing, and dispatch — and through **independent tailors** who work from their own shop or home on their own phone or tablet (a BYOD — bring-your-own-device — workforce).

The platform operates several hubs in one state with a strict **24-hour delivery promise** from the moment a garment is scanned into the hub. Today, a single customer order with three garments touches ten or more human handoffs. At each station a small piece of state has to move: paper tags, WhatsApp photos, per-station log books, spreadsheets shared on Google Drive.

Currently the journey records, at each station:

- Customer identifier (name, phone, address)
- Garment type + measurements
- Assigned hub
- Assigned independent tailor (if in stitching)
- Current stage (cut / stitch / QC / iron / packed / out / delivered)
- Date and time of every transition
- Payment status (paid online / COD)

2. The Business Problem

There is no single digital system that connects:

- Customer order → Hub intake → Tailor assignment → QC → Delivery
- Garment-level tracking across 10+ stations
- Smart matching of garments to available tailors by skill, capacity, and gender (ladies / gents / kids / unisex)
- Per-piece payout ledger for BYOD tailors
- Live "where is my order" view for the customer

As the number of hubs and customers grows, the operations team cannot answer simple questions in real time:

- How many garments are at each stage right now?
- Which tailors have capacity today?
- Which orders are at risk of breaching the 24-hour promise?
- How much has each tailor earned this week?
- Are we over-promising on tomorrow's pickup slots?
- Are COD payments being collected on delivery?

Without a single digital workflow, scaling beyond a handful of hubs is impossible — every new hub re-creates the same WhatsApp + spreadsheet tangle, and the 24-hour promise becomes a manual chase.

3. Hackathon Challenge

Build a TAILOR24-branded Digital On-Demand Tailoring & Garment Delivery Platform. The platform should connect the customer booking channel to the hub-and-tailor operations backbone and manage the complete journey from garment intake to customer delivery.

Core Workflow

Customer	Books order (address, pickup slot, garments)
Hub Intake	Garments received, QR-tagged
Cutting	Cloth cut at the hub
Independent Tailor (BYOD)	Stitches from own shop / home
Quality Check	QC pass or rework back to stitching
Ironing & Packing	Garment finished
Dispatch & Delivery	Rider delivers to customer
Customer	Receives TAILOR24 garment

4. Key Modules to Build

Module 1 — Customer Booking

Customers should be able to submit a stitching requirement through a simple web / mobile interface.

Minimum information:

- Customer name + mobile number
- Address (or pick from saved addresses)
- Garment type (shirt / trousers / kurta / suit / saree / dress / other)
- Garment gender (ladies / gents / kids / unisex)
- Service charge per garment
- Pickup slot (date + time window)
- Payment method (online / COD)

Example:

Field	Value
Name	Ravi
Mobile	98765 43210
Garments	1 shirt (gents), 1 trousers (gents)
Pickup	Tomorrow, 10:00 – 11:00
Payment	COD

Module 2 — Garment Intake & QR Tagging

When the rider picks up the order, the hub intake screen should record every garment individually and print / display a unique QR code per garment. Every subsequent station is driven by scanning this QR — no manual status fields, no paper tags.

Each garment must carry a stable identity, a customer reference, a hub assignment, and the timestamp of every transition. The event log is append-only — status is derived from the latest event, never directly written.

Module 3 — Hub Operations

Hub staff work across the canonical stages:

- Cutting — cloth cut at the hub
- Stitching — done by an independent tailor (BYOD)
- QC — pass / rework
- Ironing & packing
- Dispatch & delivery

Each station should expose a single big button / scan action — the staff member scans the garment QR and the garment advances to the next stage. A live station queue (cutting / stitching / QC / ironing / dispatch) keeps every operator focused.

Module 4 — Independent Tailor App (BYOD)

Independent tailors are the most important user. They sign up with their own phone / tablet, work from home or their own shop, and earn per-piece payouts.

The tailor app must let them:

- See garments assigned to them with QR codes
- Scan to advance a garment through stitching / rework
- Set daily availability (available / busy / on-leave)
- Share their live location so the hub can route nearby work
- Apply for leave and see their balance
- Call the customer for measurement / fabric clarification
- View earnings (earned / paid / pending) and raise a payout claim

The app must be **mobile-first and installable as a PWA** — tailors do not have company-issued devices and connectivity on the shop floor is unreliable.

Module 5 — Smart Assignment

The hub manager should be able to assign a garment to a tailor in one tap. The system should suggest the best fit using a transparent score:

- Gender match (ladies / gents / kids / unisex)
- Skill / specialisation overlap
- Daily capacity headroom (assigned today vs. capacity)
- Availability status (available / busy / on-leave)
- Rating as a tie-breaker

Example:

Garment	Best-fit tailor	Why
Ladies kurti	Lata (ladies-only, capacity 50/day)	+40 gender match, +25 skill, +10 headroom, ★4.9
Gents shirt	Santosh (active, capacity 12/day, 5 assigned today)	+40 gender match, +10 headroom, ★4.8

Module 6 — Delivery Tracking

Once a garment is packed, dispatch assigns it to a rider. The rider confirms pickup, and the customer receives a live tracking link. Suggested lifecycle:

Packed → Out for Delivery → Delivered (with OTP confirmation at the customer door). COD orders flip from *pending* to *paid* on delivery.

Module 7 — Per-Piece Payouts (Two-Step Approval)

Every completed piece credits the assigned tailor's settlement ledger. Payouts follow a two-step approval:

- Tailor raises a claim against their earned (pending) settlements
- Hub manager reviews and approves
- Admin / finance confirms the bank / UPI transfer
- Settlements flip from *pending* to *paid* with a transfer reference

This mirrors how payouts actually clear in production — the hub signs off, finance moves the money.

Module 8 — Live Dashboard

A simple dashboard should give the operations team an at-a-glance view.

Example:

Metric	Value
Garments in production	84
Garments delivered today	62
Orders at risk of missing 24h	3
Pending tailor claims	5
COD payments to collect today	12

5. Required MVP for the 24-Hour Hackathon

Teams are expected to build a working Minimum Viable Product. The minimum solution must demonstrate each of the following end-to-end with at least one happy-path walkthrough:

Customer

- View TAILOR24 information
- Submit a stitching requirement (address, garments, pickup)
- Choose online or COD payment
- Receive an order confirmation with a tracking reference

Hub Staff (cutting / QC / ironing / dispatch)

- Record garment intake (QR per garment)
- Advance a garment by scanning at each station
- View the live station queue

Hub Manager

- View all garments at the hub (men / women / kids / unisex tabs)
- View all tailors with capacity, gender, specialisation, rating
- Assign a garment to the best-fit tailor (smart suggestions)
- Approve / reject leave requests
- Approve / reject tailor payout claims
- View the live hub dashboard

Independent Tailor (BYOD)

- Sign up (account + application) and download the signed application PDF
- Set daily availability
- Share live location with the hub
- Scan garments to advance through stitching
- Call the customer for clarification
- Raise a payout claim against earned settlements

Admin / Finance

- View all hubs in the franchise
- Confirm payout transfers (the second approval step)

Rider

- See assigned pickups and deliveries
- Confirm delivery (with optional OTP at the door)

6. Innovation Opportunities

Teams may extend the MVP with additional features such as:

- Garment-level event log (every transition is an append-only record)
- 24-hour delivery countdown per order with overdue alerts
- Smart tailor assignment (gender, skill, capacity, rating)
- PWA install for tailors (BYOD — works on a personal phone over flaky connections)
- WhatsApp / SMS notifications to customers and tailors

- Per-piece payout ledger with two-step approval (manager → admin)
- Recurring daily / weekly milk-equivalent subscriptions (daily shirts, weekly kurtas)
- Demand forecasting by city / garment type
- Tailor performance analytics (rating, on-time %, rework rate)
- Customer order history + re-order with one tap
- Digital payments (UPI, cards) via any gateway with a tokenized flow
- Customer feedback and rating per garment
- Offline-tolerant PWA (queue scans, sync when back online)
- Live WebSocket push to the customer tracking page instead of polling

These are optional. Teams should prioritize a working core solution before adding advanced features.

7. Technical Freedom

Teams are free to choose their own:

- Frontend technology (web, mobile, PWA)
- Backend technology
- Database
- Cloud platform
- APIs and integrations
- UI / UX approach

The emphasis is on solving the business problem rather than using a particular technology. The repo today (Vite + React PWA, FastAPI modular monolith, Postgres) is a starting point — not a constraint.

8. Expected Outcome

The final prototype should demonstrate how TAILOR24 can move from a largely manual, WhatsApp-driven workflow to a single digital workflow:

Book → Intake → Tag → Assign → Stitch → QC → Iron → Pack → Dispatch → Deliver → Track → Pay

The solution should be:

- Simple — usable by a first-time tailor on their own phone
- Mobile-friendly — works on a small screen over a 3G connection
- Trustworthy — every garment has a real audit trail
- Modular — designed for one hub or seven, one tailor or seventy
- TAILOR24-branded — feels like the product, not a hack

9. Final Challenge Statement

“How can we build a single digital platform that connects TAILOR24 customers, hubs, and independent tailors — tracking every garment individually from booking to delivery under a 24-hour promise, with smart assignment, BYOD-friendly mobile apps, and per-piece payouts to the workforce?”

Book it. Tag it. Stitch it. Pay it.

Duration: 24 Hours